

INFO8006: Project 2 - Report

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1 Bayes filter

a. Sensor model. Let F be the traversable cells, X_t the ghost position and q_t the known Pacman position. Let $d(x,q)$ be Manhattan distance. For requested variance v , $p=1/2$ and $n=\text{floor}(4v)$. The measurement is a distance plus independent centered binomial noise:

$$E_t = d(X_t, q_t) + B_t - np, \quad B_t \sim \text{Binomial}(n, p)$$

For $k=e-d(x,q_t)+np$, the likelihood is the following expression when k is an integer from 0 to n , and zero otherwise:

$$P(E_t = e \mid X_t = x, q_t) = \binom{n}{k} p^k (1-p)^{n-k}$$

The conditional mean is $d(x,q_t)$, and the actual variance is $n/4$, which can be below v due to flooring. At $v=1$, the noise values $-2,-1,0,1,2$ have probabilities $1,4,6,4,1$ divided by 16. Odd n gives half-integer noise. Negative measurements are possible and must not be clipped. At $n=0$, the sensor is exact.

b. Unified transition model. Let $N(x)$ be the traversable orthogonal neighbors of x . With Pacman fixed at q during a ghost move, assign weight λ to neighbors y with $d(y,q) \geq d(x,q)$, and weight 1 otherwise. Normalize these weights over $N(x)$:

$$T_\lambda(y \mid x, q) = \frac{w_\lambda(y, x; q)}{\sum_{z \in N(x)} w_\lambda(z, x; q)}, \quad y \in N(x)$$

The single free parameter λ equals 1 for confused, 2 for afraid and 8 for scared. Other destinations have zero probability. All neighbors are eligible, including the previously visited cell. There is no direction memory. Staying still is excluded when a neighbor exists; an isolated cell has a self-transition of probability 1.

2 Implementation

a. Leave empty.

3 Experiment

a. Uncertainty. We use Shannon entropy in bits for each live ghost; lower values mean a more concentrated belief. Terms with zero probability contribute zero. Eaten ghosts are excluded.

$$H(b_t) = - \sum_{x \in F} b_t(x) \log_2 b_t(x)$$

b. Quality. With true ghost position x^* , we use expected Manhattan distance from belief to truth. Smaller is better. This penalizes mass far from truth, whereas entropy alone does not establish correctness. Ground truth is used for evaluation:

$$L(b_t, x^*) = \sum_{x \in F} b_t(x) d(x, x^*)$$

c. Experimental results. We ran 180 trials: 2 layouts x 3 policies x variances $\{0.25, 1, 4\}$ x 10 seeds (0-9). Each trial used one ghost and lasted all 200 observations. The same Pacman random walker was used throughout. It reads truth to avoid collisions; the filter does not. This controller affects the observation trajectory, so these results are conditional on this protocol.

Curves show trial means and pointwise 95% Student-t confidence intervals across trials. For bars and the variance sweep, each trial contributes one mean over $t=100-199$; confidence intervals use the ten independent trial means, rather than treating time steps as independent samples. These are late-window summaries, not a guarantee of stationarity.

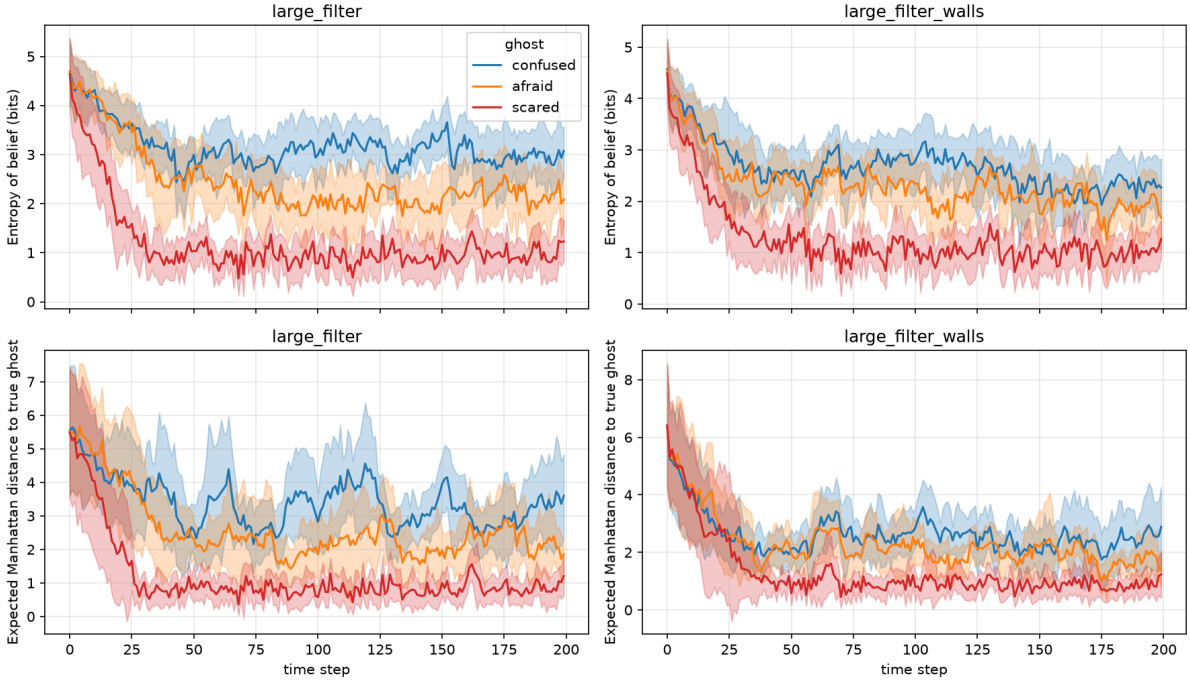


Figure 1. Default variance 1: entropy and localization error over time, with shaded 95% confidence intervals.

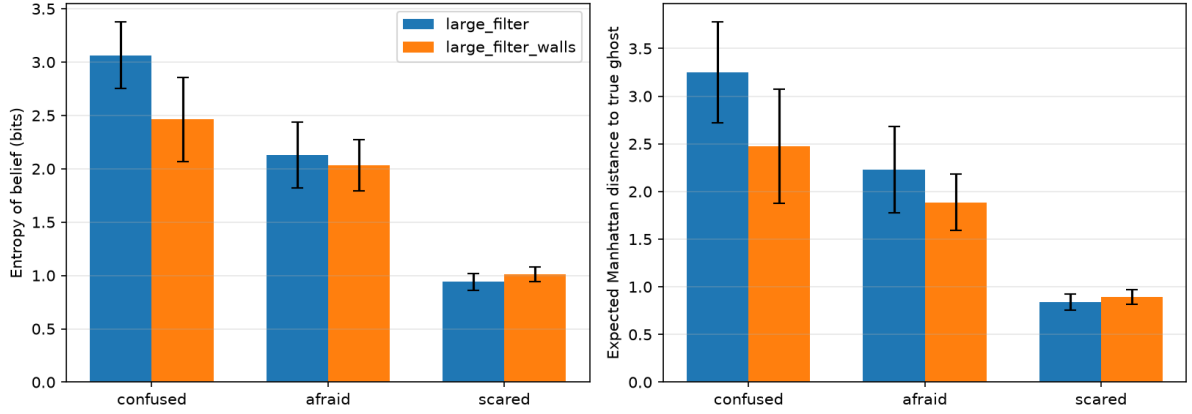


Figure 2. Default variance 1: late-window means and 95% CI.

At variance 1, entropy/error on `large_filter` are 3.063/3.248 (confused), 2.129/2.229 (afraid), 0.939/0.838 (scared). On `large_filter_walls` they are 2.462/2.473, 2.033/1.886 and 1.009/0.894 respectively. Exact confidence intervals are in the accompanying summary.csv.

Convergence check. Initial 200-step trials showed continued entropy reduction for confused on `large_filter_walls`: differences between $t=150-199$ and $t=100-149$ were -0.428 ± 0.288 bits at $v=1$ and -0.262 ± 0.195 at $v=4$ (paired 95% CI). We therefore extended these two conditions to 1,000 steps using the same ten seeds. All 20 extended trials reached 1,000 steps. Comparing $t=750-999$ with $t=500-749$, entropy drift was -0.148 ± 0.143 bits at $v=1$ and -0.167 ± 0.117 bits at $v=4$ (paired 95% CI). Thus some drift remains even at 1,000 steps; we do not claim convergence. Neither a small drift nor a confidence interval containing zero proves convergence; larger sample sizes may still be needed. The 10-trial baseline is a limitation of this report.

d. Transition parameter. With a non-decreasing-distance neighbors and b decreasing-distance neighbors, escape probability is $\lambda a / (\lambda a + b)$. For $a=b=1$, this is $1/2$, $2/3$ and $8/9$. A higher parameter strengthens escape preference. If all legal neighbors belong to the same group, the parameter has no effect; dead ends can force motion toward Pacman.

At default sensor variance, scared has the lowest entropy and error in both layouts, consistent with a more constrained escape distribution. Walls alter legal transitions and can help constrain possible locations, but confused/afraid improve on the walls layout while scared worsens slightly. Thus neither more fear nor more walls guarantees an improvement in every setting. Differences in means alone are not evidence of statistical significance.

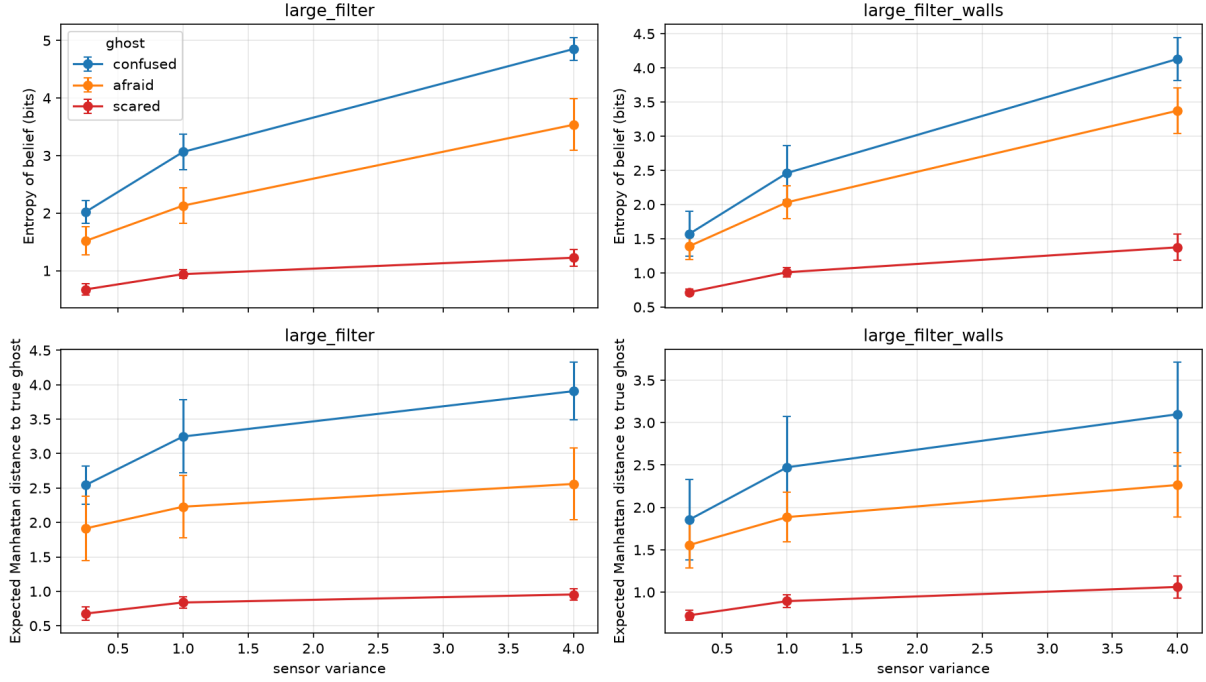


Figure 3. Sensor variance sweep, late-window means +/- 95% CI.

e. Sensor variance. For every layout/policy pair, increasing variance from 0.25 to 1 to 4 increases mean entropy and expected distance error. Broader noise makes a measurement compatible with more positions, weakening correction. Transition information still matters: scared remains easier to localize in these trials. The chosen variance levels equal the actual $n/4$ values; arbitrary inputs can be rounded down.

f. Belief-only controller. Exclude zero-mass beliefs for eaten ghosts. Select the remaining ghost with the smallest expected Manhattan distance from Pacman, then choose a legal non-STOP action whose successor minimizes that expected distance. Ties follow legal-action order; if no target or move exists, return STOP. The controller uses only current position, legal actions and belief, not true ghost positions. Greedy Manhattan distance can oscillate around walls, so it does not guarantee eventual capture.

Bonus smoke tests at seed 0, variance 1 and one ghost won all six layout/policy pairs within a 100-step cap: 15/45/27 steps on large_filter and 12/16/72 on large_filter_walls for confused/afraid/scared. This is a smoke test, not a multi-seed success-rate estimate. Separate filter integration tests with three afraid ghosts completed 60 steps for seeds 0 and 1.

g. Leave empty.